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EE321: Computer Architecture
Control Solution (1h)
2024 – 2025

Notes: Answer briefly and clearly using the provided space. No extra sheet will be accepted.

1. Z80 CPU Function and Interrupt Processing (20 pts)

Consider Z80 CPU with the following specifications: little-endian 8-bit processor, 16-bit address bus, a partial list of instructions and opcodes is given in Table 1, the instruction formats are given in Table 2. The programming model is given in Table 3 and the GPR codes are given Table 4. The number of clock cycles for each Z80 machine cycle are given in Table 5. Consider the following HEX memory content:

2340H: 11 F1 12 F2 13 F3 14 F4 15 F5 16 F6 17 F7 18 F8
F610H: 01 F1 02 F2 03 F3 04 F4 05 F5 06 F6 07 F7 08 F8

1.1 Give the size of the registers A, MAR, MBR, and IR.

A and MBR: 8 bits
MAR: 16 bits
IR: 32 bits *(1,5 pt.)*

1.2 Give in decimal the 2'C 8-bit value stored in memory locations: a) 2348H and b) 2349H.

a) [2348H] = $(15H)_{2'C}$ = 21
b) [2349H] = $(F5H)_{2'C}$ = -11 *(1,5 pt.)*

1.3 Give in decimal the 2'C 16-bit value stored in memory locations: a) 2348H and b) 2349H.

[2348H] = $(F515H)_{2'C}$ = -2795
[2349H] = $(16F5H)_{2'C}$ = 5877 *(1,5 pt.)*

1.4 Give the size in bytes of the instructions: a) `LD IX,nn`, b) `LD HL,(nn)`, and c) `LD R,S`.

a) LD IX, nn: 4 bytes.
b) LD HL, (nn): 3 bytes.
c) LD R,S: 1 byte. *(1,5 pt.)*

1.5 Give the HEX code of the instructions: a) `ADD A,-1`, b) `ADD A,(HL)`, c) `ADD A,H`.

a) ADD A,-1: C6FFH
b) ADD A,(HL): 86H
c) `ADD A,H`: 10000100 = 84H *(1,5 pt.)*

1.6 Give the HEX content of the registers PC, MAR, and MBR after executing the **fetch** cycle of the instruction `LD IX,(2348H)` stored at address 2100H.

The HEX code of the instruction is: DD2A4823H stored at address 2500H. After the opcode fetch machine cycle, we have:
PC = 2104H
MAR = 2103H
MBR = 23H *(1,5 pt.)*

1.7 Give the HEX content of the registers PC, MAR, and MBR after executing the **execute** cycle of the instruction `LD IX,(2348H)` stored at address 2100H.

The HEX code of the instruction is: DD2A4823H stored at address 2100H. After the execute cycle, we have:
PC = 2104H
MAR = 2349H
MBR = F5H *(1,5 pt.)*

1.8 Which of the Z80 machine cycles (see Table 5) are involved in the instructions: a) `LD IX,nn`, b) `LD HL,(nn)`, and c) `LD R,S`?

a) `LD IX,nn:` opcode fetch, operand fetch.
b) `LD HL,(nn):` opcode fetch, operand fetch, memory read.
c) `LD R,S:` opcode fetch. *(1,5 pt.)*

1.9 Give the number of clock cycles (T-sates) it takes to execute (i.e. fetch and execute) the instructions: a) `LD IX,nn`, b) `LD HL,(nn)`, and c) `LD R,S`.

a) `LD IX,nn:` 14 clock cycles (8 for opcode fetch, 6 for operand fetch).
b) `LD HL,(nn):` 16 clock cycles (4 for opcode fetch, 6 for operand fetch, and 6 for memory read).
c) `LD R,S:` 4 clock cycles (4 for opcode fetch). *(1,5 pt.)*

1.10 Assuming SP=F612H, give the HEX content of the registers SP, H, L, MAR, and MBR after execution the instruction `POP HL`.

SP = F614H
MAR = F613H
MBR = F2H
H = F2H
L = 02H *(1,5 pt.)*

1.11 We assume Z80 is configured in interrupt mode 2, register I=23H, and SP= F612H. The CPU receives an interrupt during the execution of the instruction `LD IX,0H` stored at address 2100H. The provided data by the interrupted device is 4CH. Give the HEX content of SP and PC **before** executing the **interrupt cycle.**

SP = F612H
PC = 2104H *(1,5 pt.)*

1.12 We assume the configuration of question 1.11, give the HEX content of SP and PC **after** executing the **interrupt cycle.**

SP = F610H
PC = [234CH] = F717H *(1,5 pt.)*

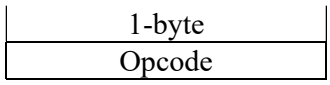
1.13 We assume the configuration of question 1.11, give the HEX content of IR, MAR and MBR before fetching the first instruction of the ISR.

IR = DD210000H
MAR = 234DH
MBR = [234DH] = F7H *(2 pts.)*

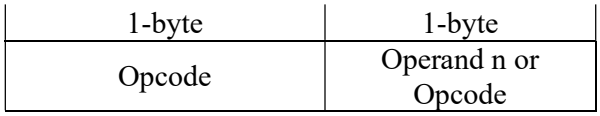
Instruction	Opcode (binary or HEX)	Instruction Size	Description
LD R, S	$01R_2R_1R_0\ S_2S_1S_0$		R= S
LD IX, nn	DD21H		IX = nn
LD IX, (nn)	DD2AH		IX = [nn]
LD HL, nn	21H		HL = nn
LD HL, (nn)	2AH		HL = [nn]
LD R, (HL)	$01\ R_2R_1R_0110$		R = [HL]
LD (HL), S	$01110\ S_2S_1S_0$		[HL] = S
LD (nn), A	32H		[nn] = A
PUSH IX	DDE5H		SP = SP-2, [SP] = IX
POP IX	DDE1H		IX=[SP], SP = SP+2
PUSH HL	E5H		SP = SP-2, [SP] = HL
POP HL	E1H		HL=[SP], SP = SP+2
ADD A, n	C6H		A = A + n
ADD A, (HL)	86H		A = A + [HL]
ADD A, S	$10000S_2S_1S_0$		A = A + S
INC (HL)	34H		[HL] = [HL] + 1
INC R	$00R_2R_1R_0100$		R = R + 1
HALT	76H		Stop the execution of the program

R =Destination register (it can be any GPR register).
S = Source register (it can be any GPR register).
$R_2R_1R_0$ = Binary code of the register R (see Table 4).
$S_2S_1S_0$ = Binary code of the register S (see Table 4).
n =8-bit value.
nn =16-bit value.
[…] = Content of memory location.

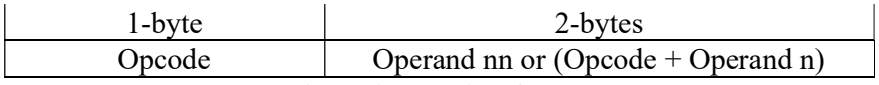
Table 1: Partial list of Z80 instructions.



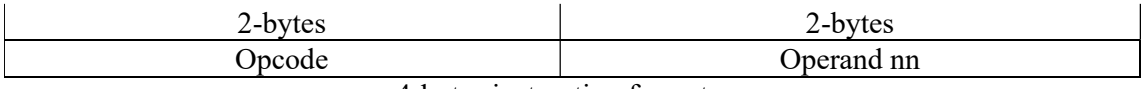
1-byte instruction format



2-bytes instruction format.



3-bytes instruction format.



4-bytes instruction format

Table 2: Z80 instruction formats.

Register names	Description
A	Accumulator
B, C, D, E, H, and L	8-bit general purpose registers (GPR), can also be used in pairs as 16-bit registers (i.e. BC, DE, HL)
IX, IY	16-bit Address registers
F	8-bit flag register
I	8-bit interrupt register
R	8-bit refresh register
PC	Program counter
SP	Stack pointer

Table 3: Z80 programming model.

GPR	A	B	C	D	E	H	L	
Code	111	000	001	010	011	100	101	

Table 4: Binary codes of Z80 accumulator and GPR.

Z80 machine cycle	Number of clock cycles (T-states)
1-byte opcode fetch	4 clock cycles
1-byte operand fetch	3 clock cycles
1-byte memory read	3 clock cycles
1-byte memory write	3 clock cycles
1-byte I/O read	4 clock cycles
1-byte I/O write	4 clock cycles

Table 5: Number of clock cycle for each Z80 machine cycle.

Z80 interrupt signal	Mode	ISR location
Reset		0000H
NMI (Non-Maskable int.)		0066H
INT (Maskable int.)	0	ISR location can be one of the eight memory locations 00H, 08H, 10H, … 38H
	1	0038H
	2	8-bit data is supplied by the interrupting device. The ISR address is: [(I register * 256) + 8-bit data]

Table 6: Z80 interrupt processing.